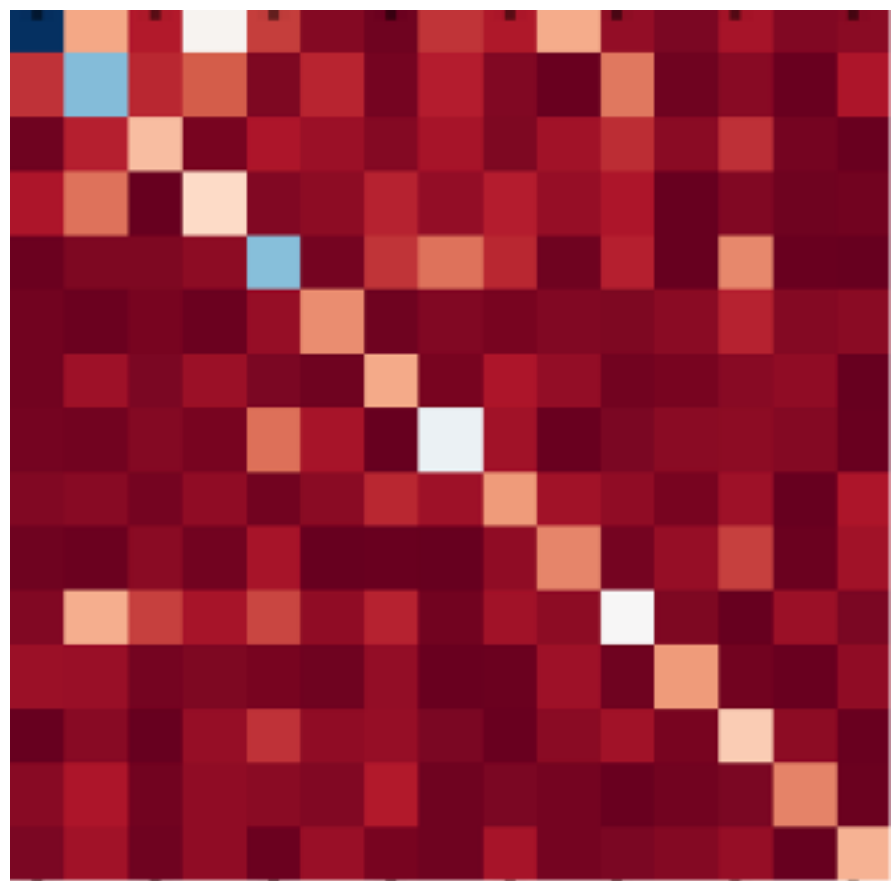
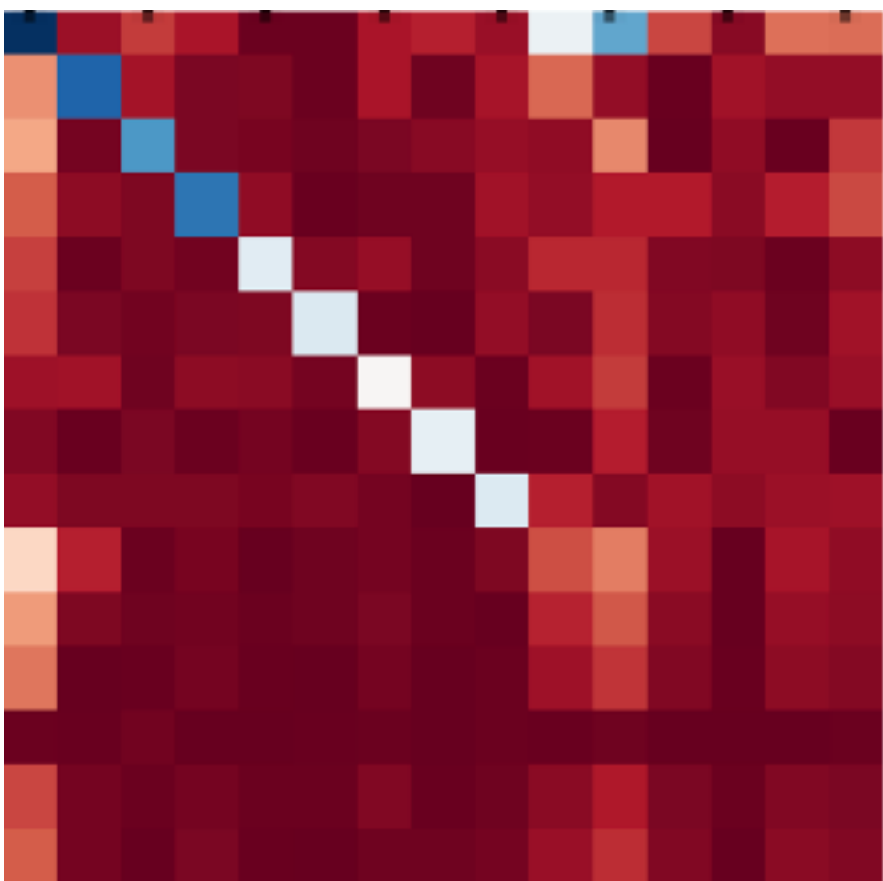
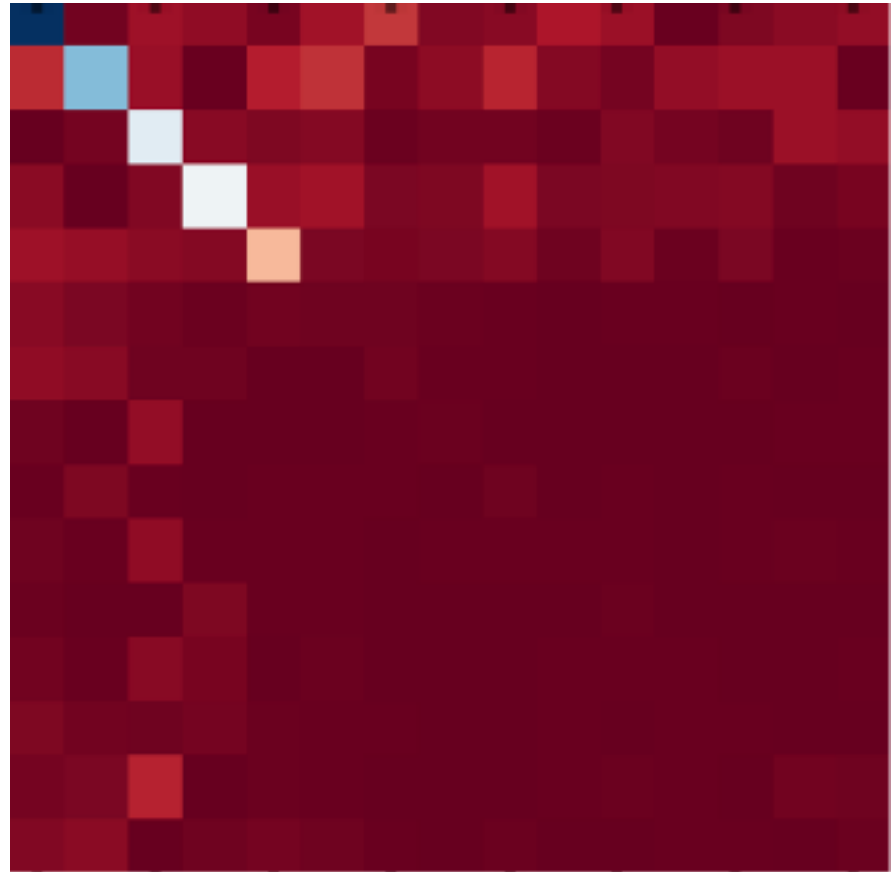
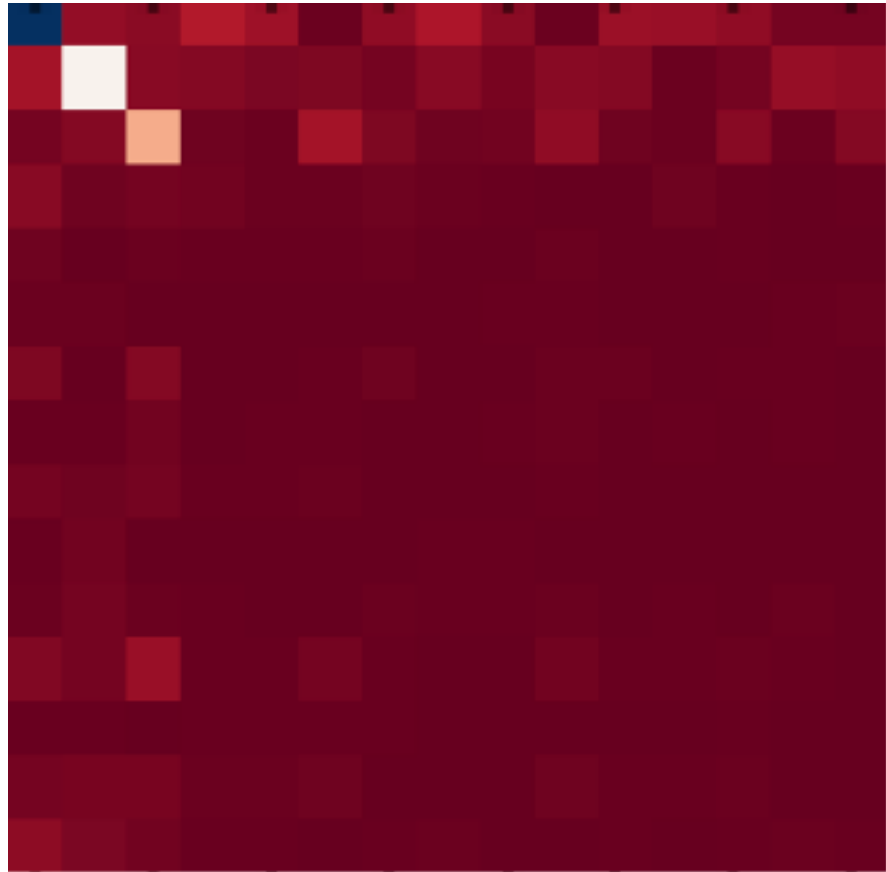
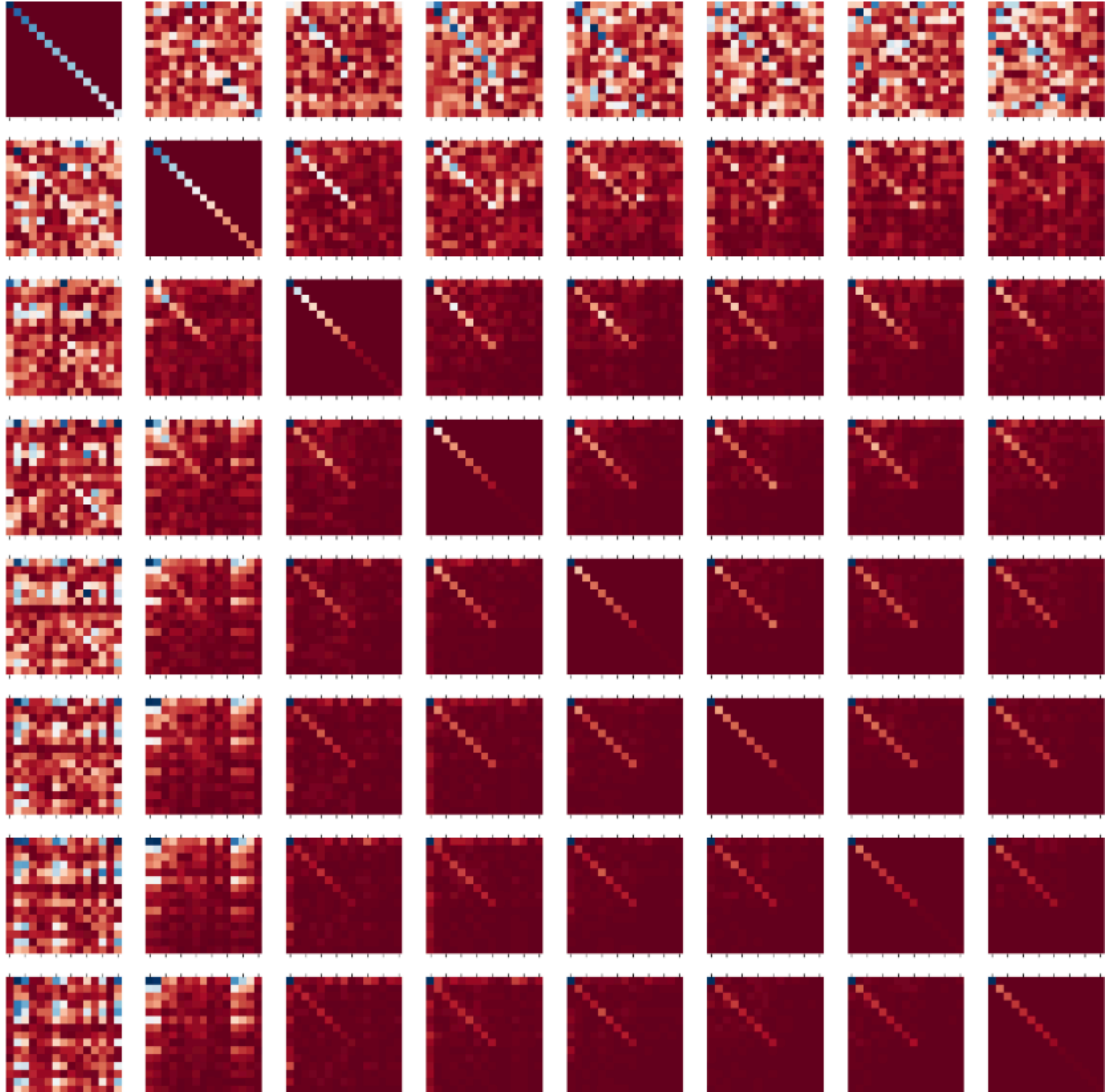
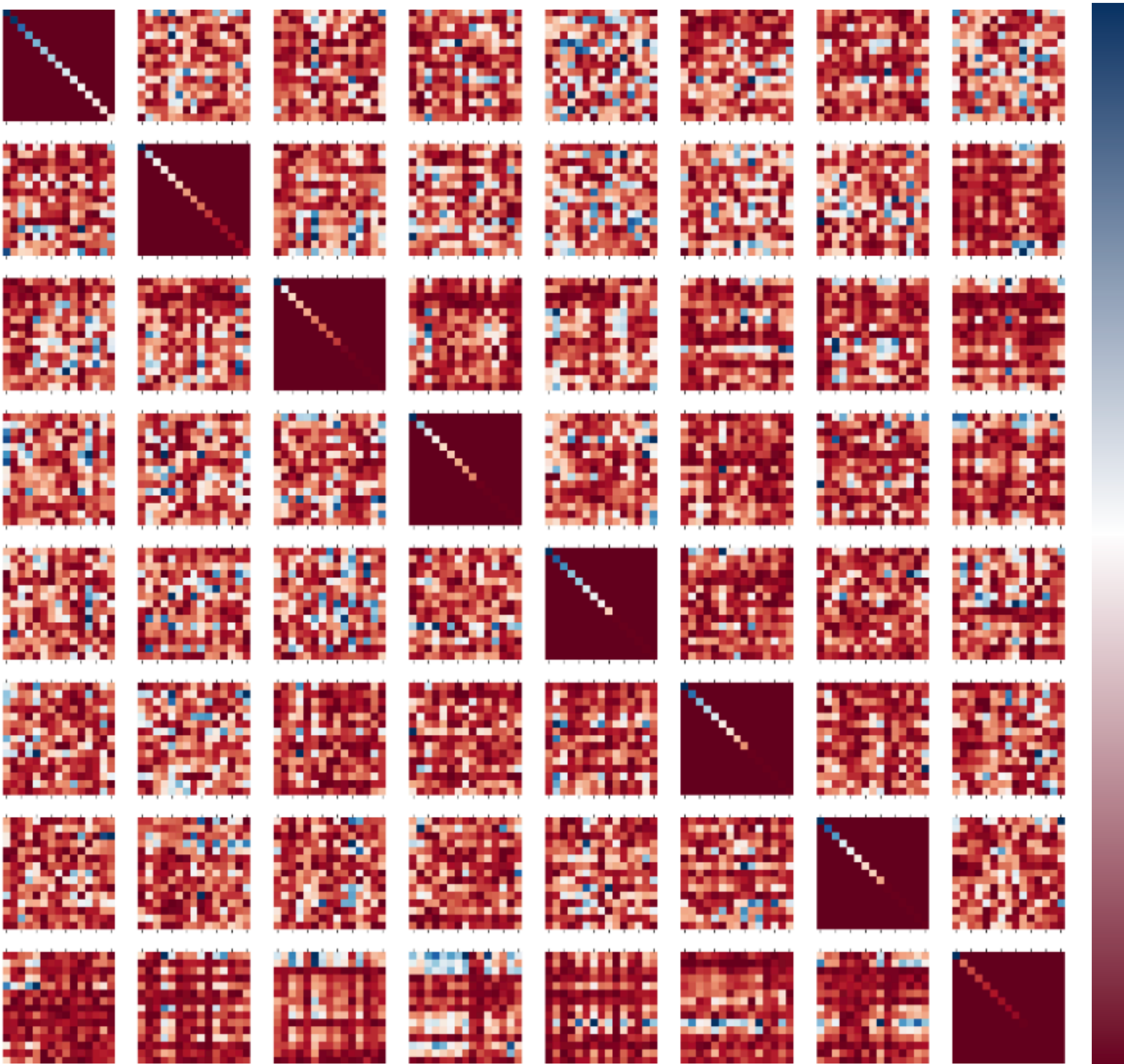




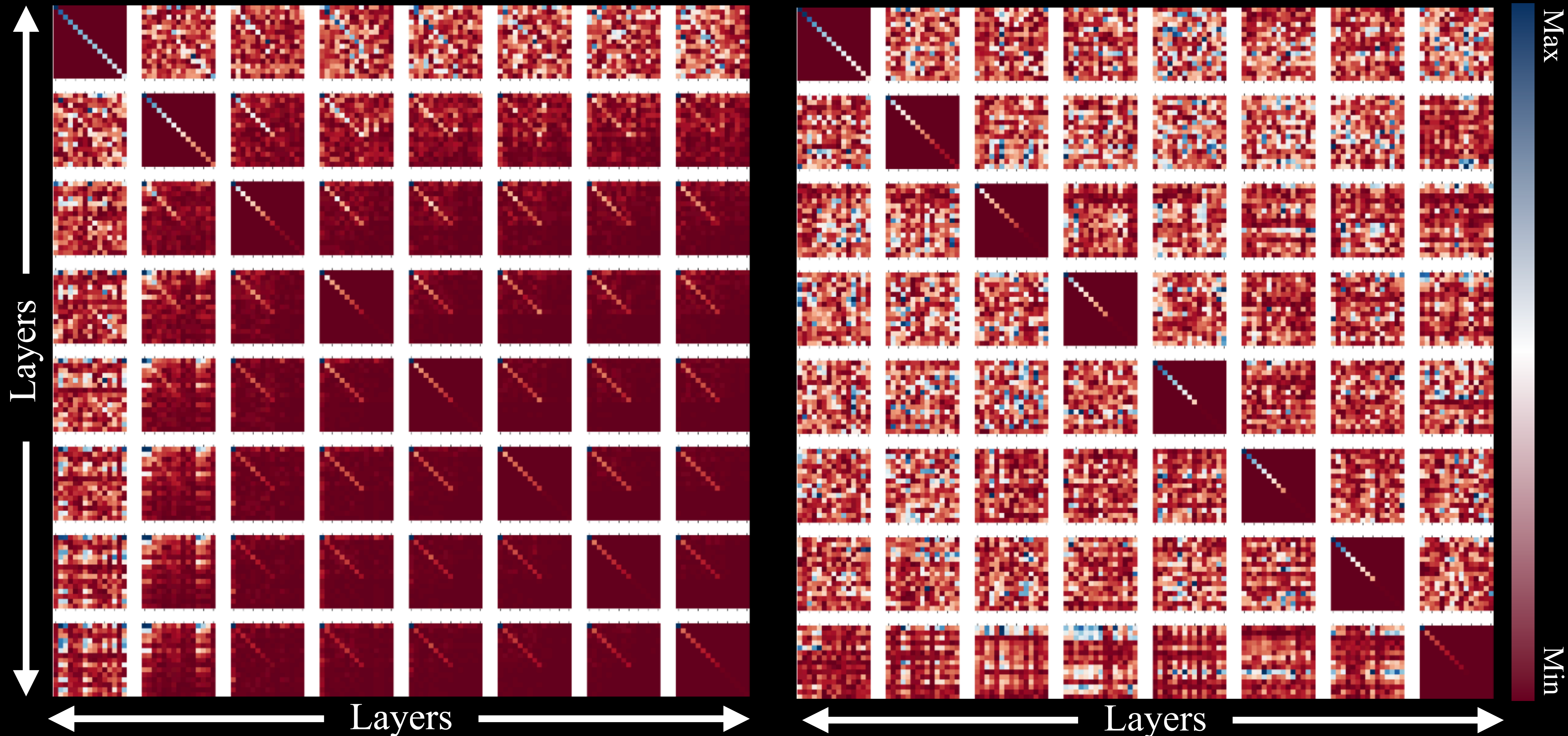






WITH Skip Connections

WITHOUT Skip Connections



ResNet34 trained on CIFAR10

Unconstrained Jacobians

$$\min_{w, \{J_l\}_{l=1}^L} \mathcal{L} \left(w^T \prod_{l=1}^L (I + J_l) X, y \right) + \frac{\lambda}{2} \sum_{l=1}^L \|J_l\|_F^2 + \frac{\lambda}{2} \|w\|_2^2$$